1. Architecture nima?

- Bitta tizimning ichki tuzilishi, uning qismlari va bu qismlar bir biri bilan qanday bog'langanligini anglatadi.
- AI da Architecture deganda model qanday tuzilganligi tushuniladi.

Masalan: Neural Network architectures
- nechta layer borligi
- har bir layer qanday vazifa bajarishi.
- layerlar qanday tartibda joylashgani
- ular bir-biri bilan qanday bog'langanligi

Input -> Hiddin layer -> output va

2. Supervised ML Architecture.

Dataset — idea, data topish. features, target.

↓

Data preprocessing — missing values, encoding, scaling.

↓

Train test split — Data leakageni oldini olish uchun preprocessingdan oldin qilish kerak.

↓

Model — Linear Regression, Decision tree...

↓

Evaluation — Regression: MAE, RMSE, R^2. Classification: Accuracy, Precision, Recall, F1.

↓

Prediction.

3. Image Classification ni qayerlarda ko'rishimiz mumkin. (Real hayotlarda).

① Agriculture - o'simlik kasalligini aniqlash
Model barg rasmini Healthy, Early Blight yoki Late Blight kabi kategoriyalarga ajratadi

Multiclass

② Healthcare: tibbiy tasvirlarni tasniflash.

- Model tasvirlarni normal yoki Pneumonia suspected kabi sinflarga ajratadi.

Binary

4. Computer Vision model qilishdagi qiyinchiliklar:

① Data: Yetarli, ishonchli va xilma xil datalarni toplash qiyin.

② Computational resources
GPU, xotira va training vaqti katta xarajat

③ Generalization va Robustness.
Model yangi sharoitlarda, masalan: boshqa yorug'lik yoki kamera ~~burchak~~ har xil burchaklardan olinganda ham ishonchli ishlashi kerak

④ Data class imbalance.
Bazi sinflarda ko'p, boshqasida kam bo'lsa, model ko'p uchraydigan sinflarga og'ib ketishi m-n.